AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 20 · STUDENT LESSON

# Conversation State and Memory

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

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## Before you start

AI-2 Week 4: state records the current values.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Explain the mechanism and terminology of conversation state and retention. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of conversation **state** and retention.

I can apply the **state**d procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 20: Conversation **State** and Memory

Products choose what context to keep, summarize, forget, or retrieve.



CREATOR MISSION

Design memory rules for a tutoring conversation.

Creator name and date:

> **Write or draw your reasoning and evidence.**

### Readiness for the connected explanation

Recall Tools and API contracts. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

### This week's end goal

By the end, I can explain conversation **state** and memory using a traced example and a **state**d limitation.

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STUDENT LESSON · 2 OF 18

## Notice the evidence

> Investigate the supplied Conversation State and Memory case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



### Start with this puzzle

MYSTERY

The learners must use conversation **state** and memory to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

### A closer explanation

Conversation **state** stores facts or prior interactions for later use. It needs scope, provenance, correction and **retention rule**s rather than indefinite accumulation.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

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STUDENT LESSON · 3 OF 18

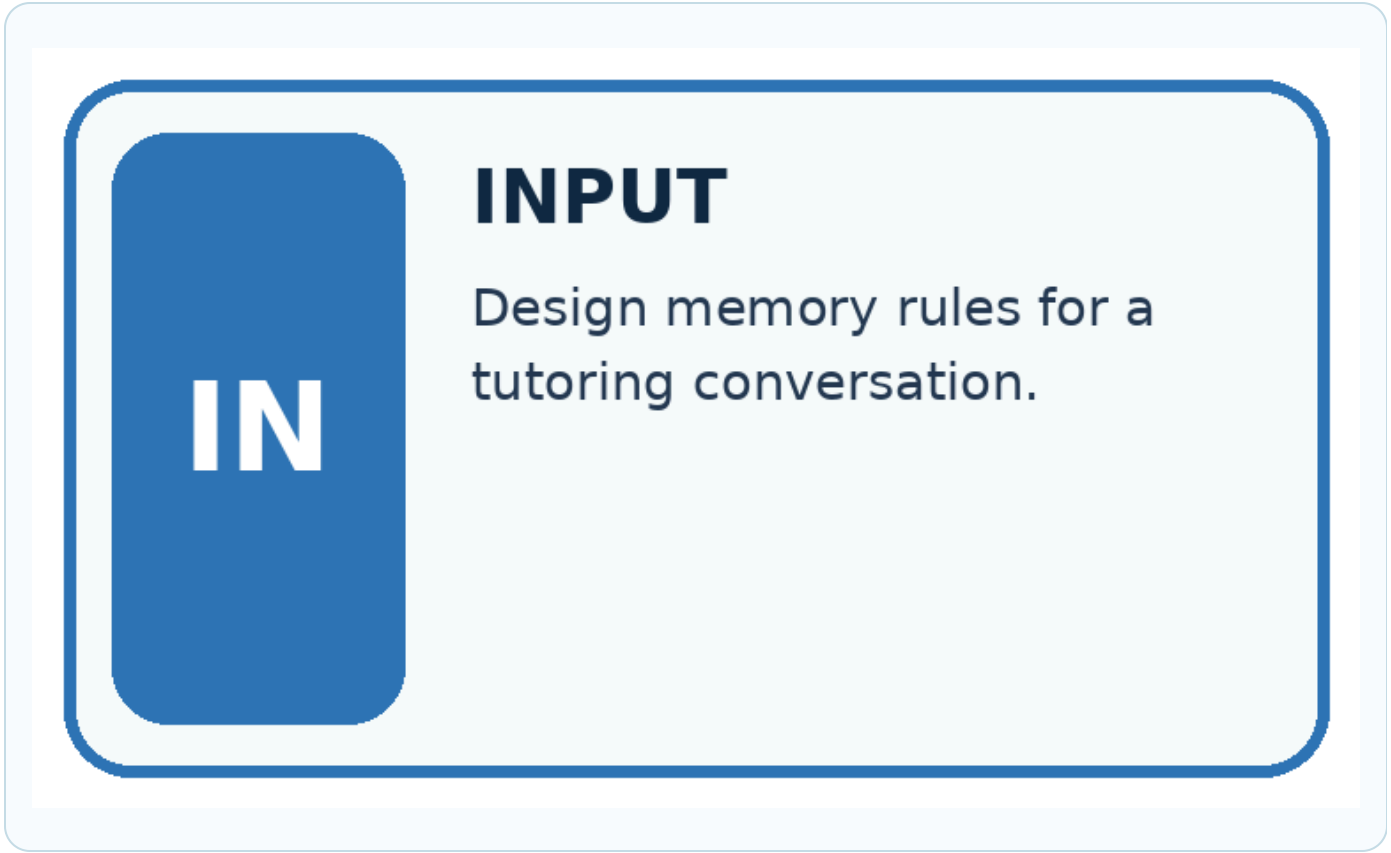
## See how it works

> Follow the mechanism for Conversation State and Memory. Connect each labelled input, deciding step and result.

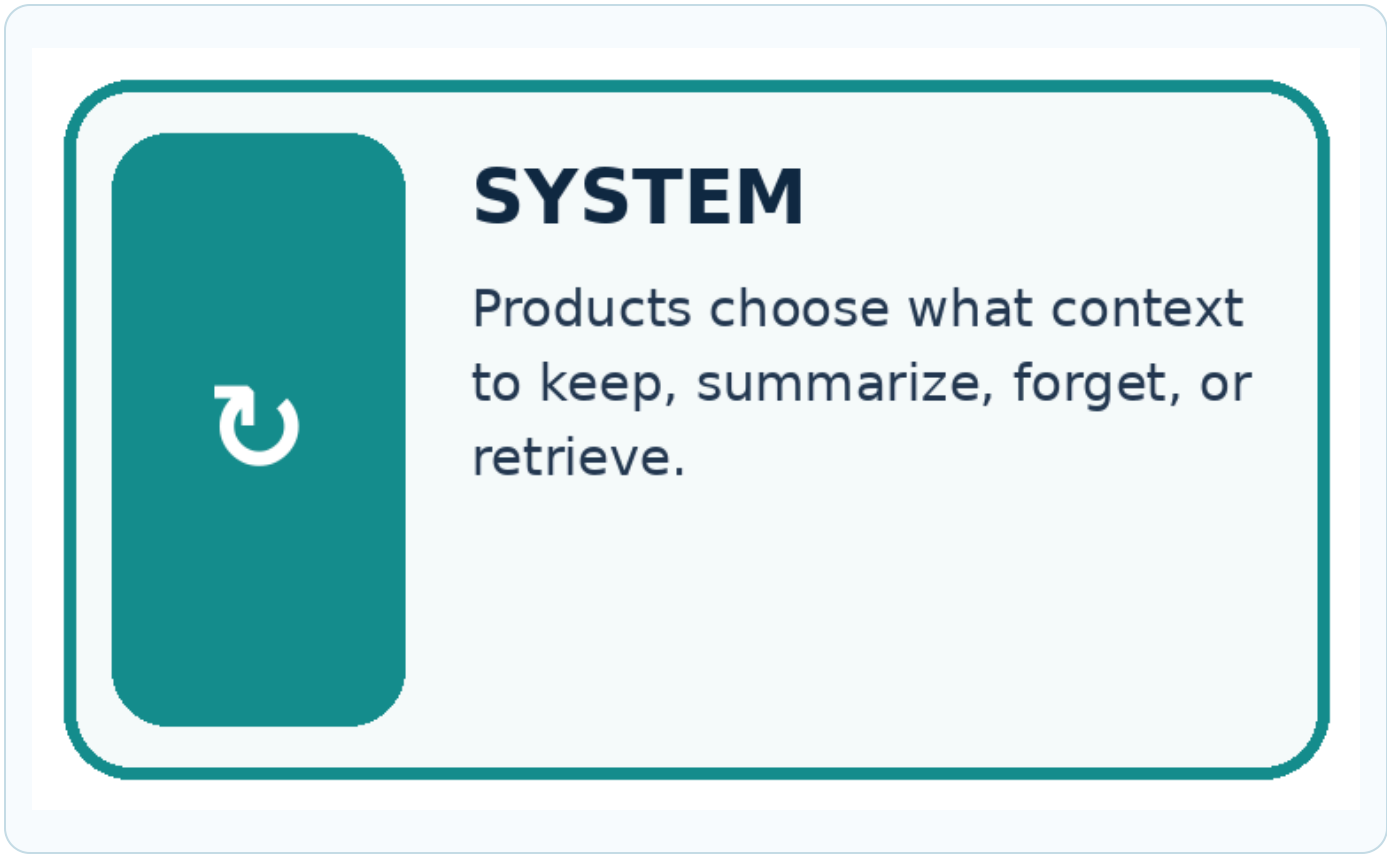
VISUAL MECHANISM

**Input → Process → Output**

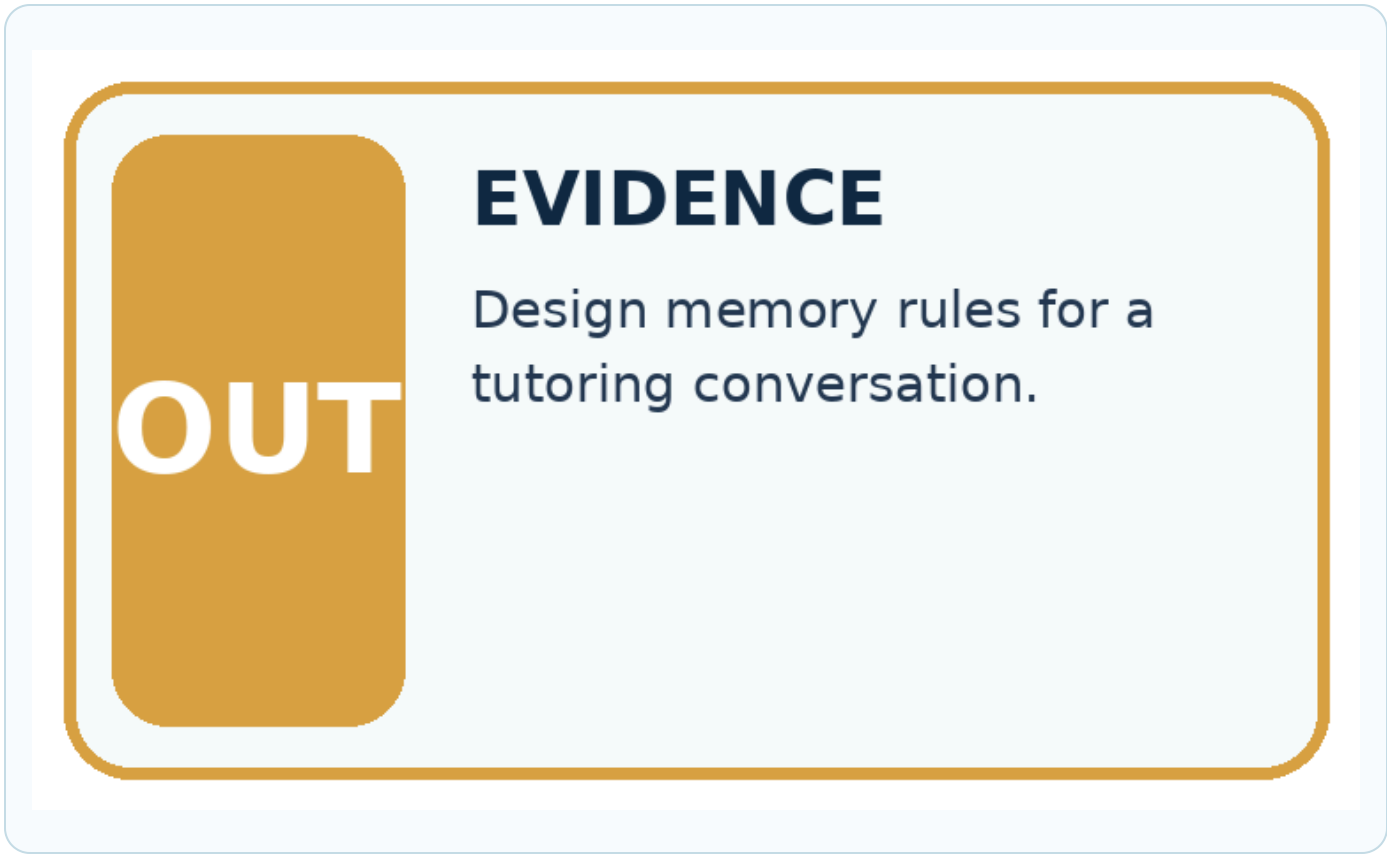
Trace the information. At each stage, ask what changed and how you could test it.




↓


SYSTEM
Products choose what context to keep, summarize, forget, or retrieve.

↓


OUT
EVIDENCE
Design memory rules for a tutoring conversation.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

### A closer explanation

This classroom assistant keeps an approved room ID only for the current session. A later correction replaces the active value but retains an audit event until session end.

Do not persist private text merely because it appeared in conversation.

**Conversation state and retention — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary. | This classroom assistant keeps an approved room ID only for the current session. A later correction replaces the active value but retains an audit event until session end. Do not persist private text merely because it appeared in conversation. | Active state should be B under the correction rule. The summary is stale; regenerate or invalidate it and preserve the correction audit within the declared session lifetime. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the **state**d contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

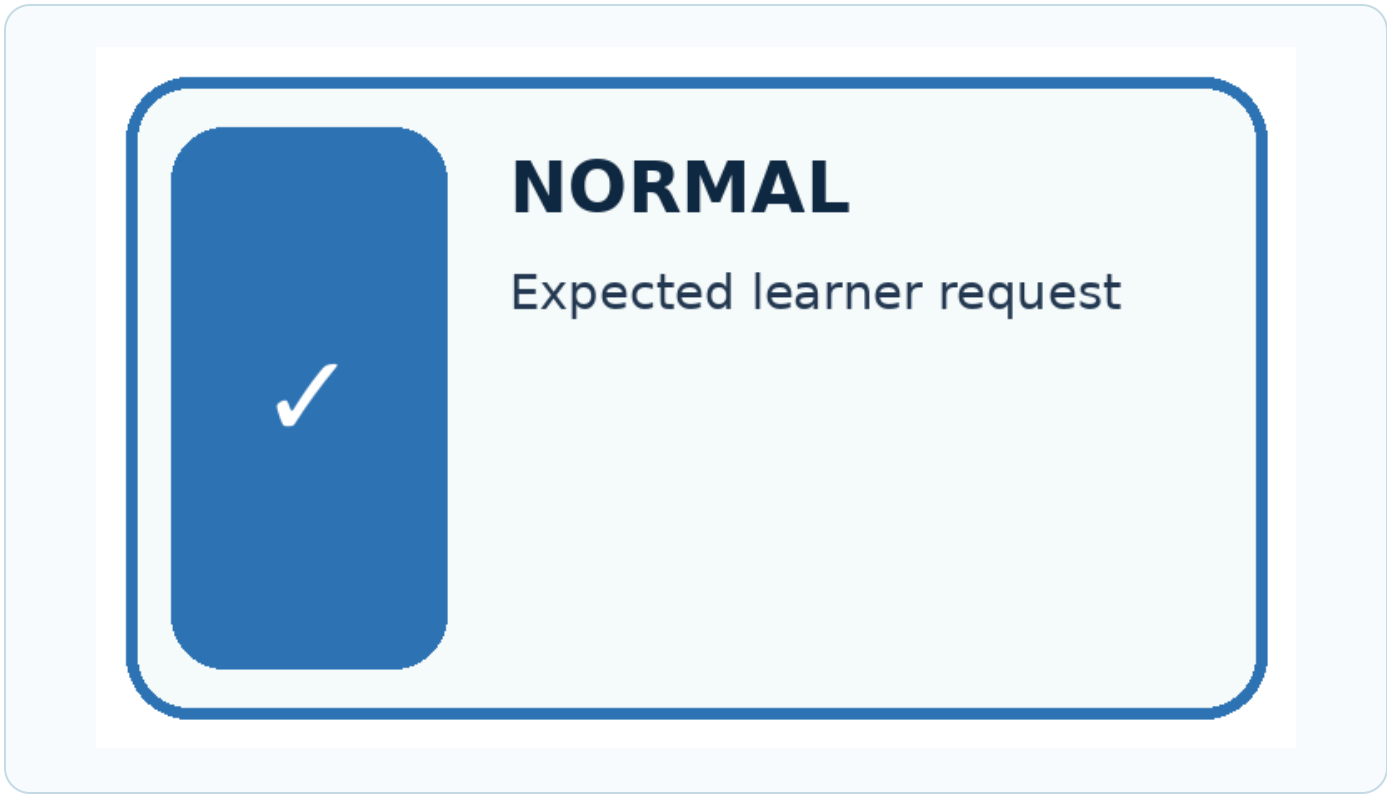
> Compare the supplied cases for Conversation State and Memory. Identify the detail that changes the conclusion.

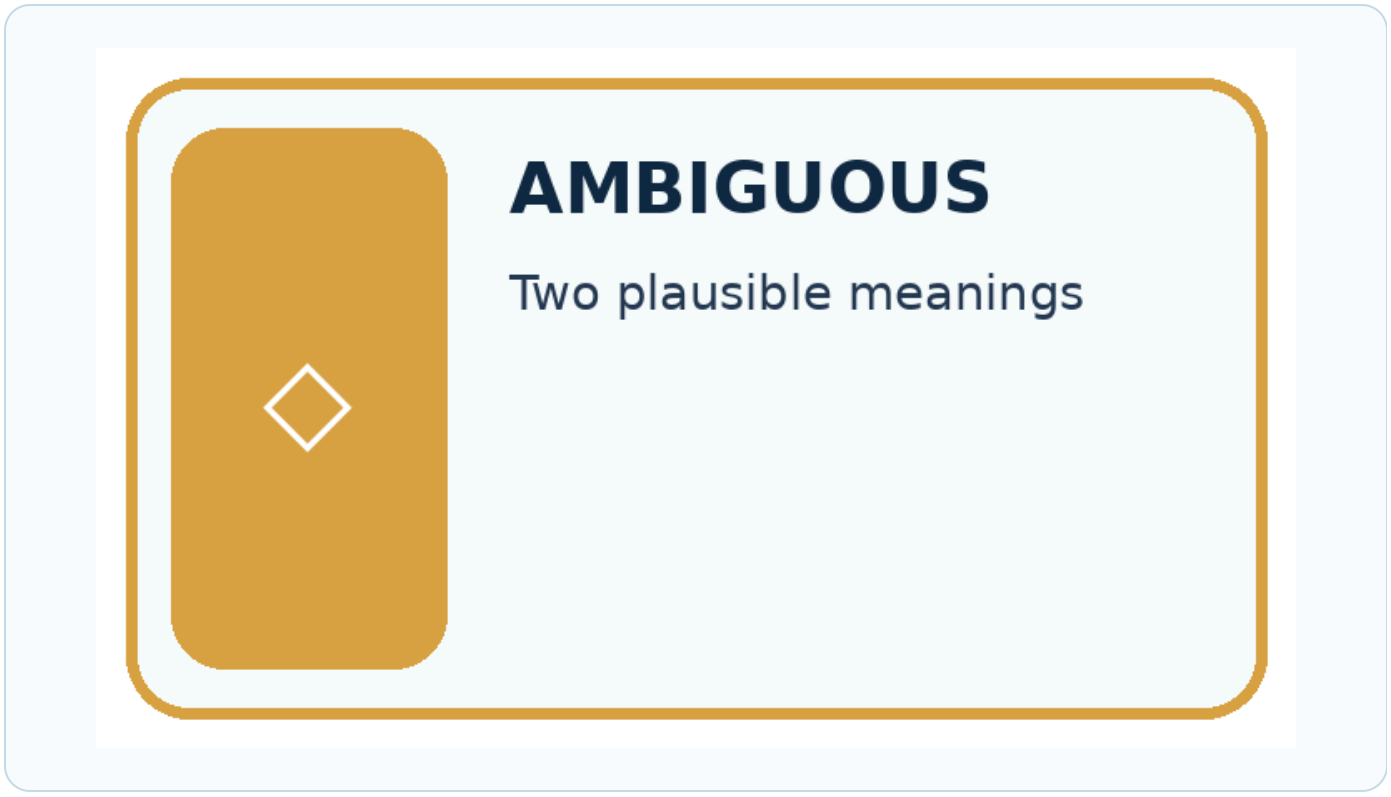
### The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

### The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.






### Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

**Change one thing in the pictured case**

### Change one thing

Change: The next session asks “summarise my book” without naming one.

### Predicted result

**Prediction**: Ask which book the learner wants.

### Why the result follows

Reason: The declared policy clears **session state**, so yesterday’s choice is unavailable.

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STUDENT LESSON · 5 OF 18

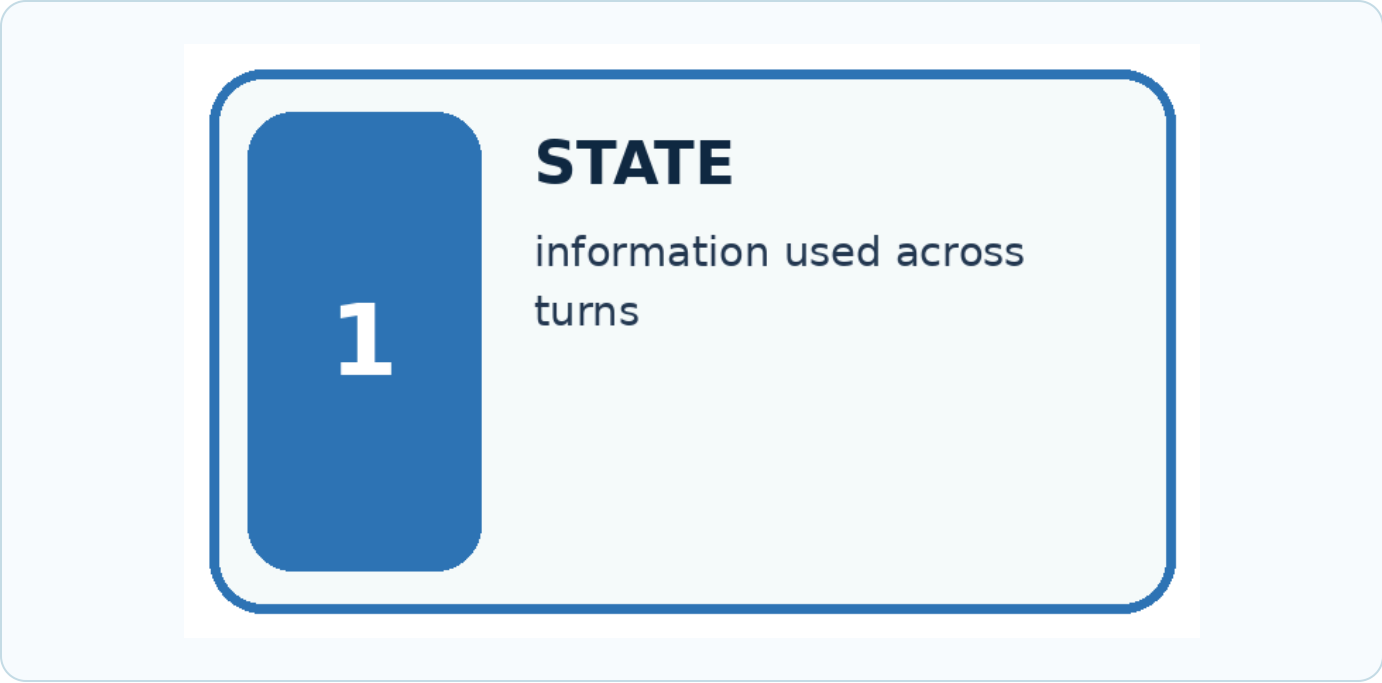
## Words that unlock the idea

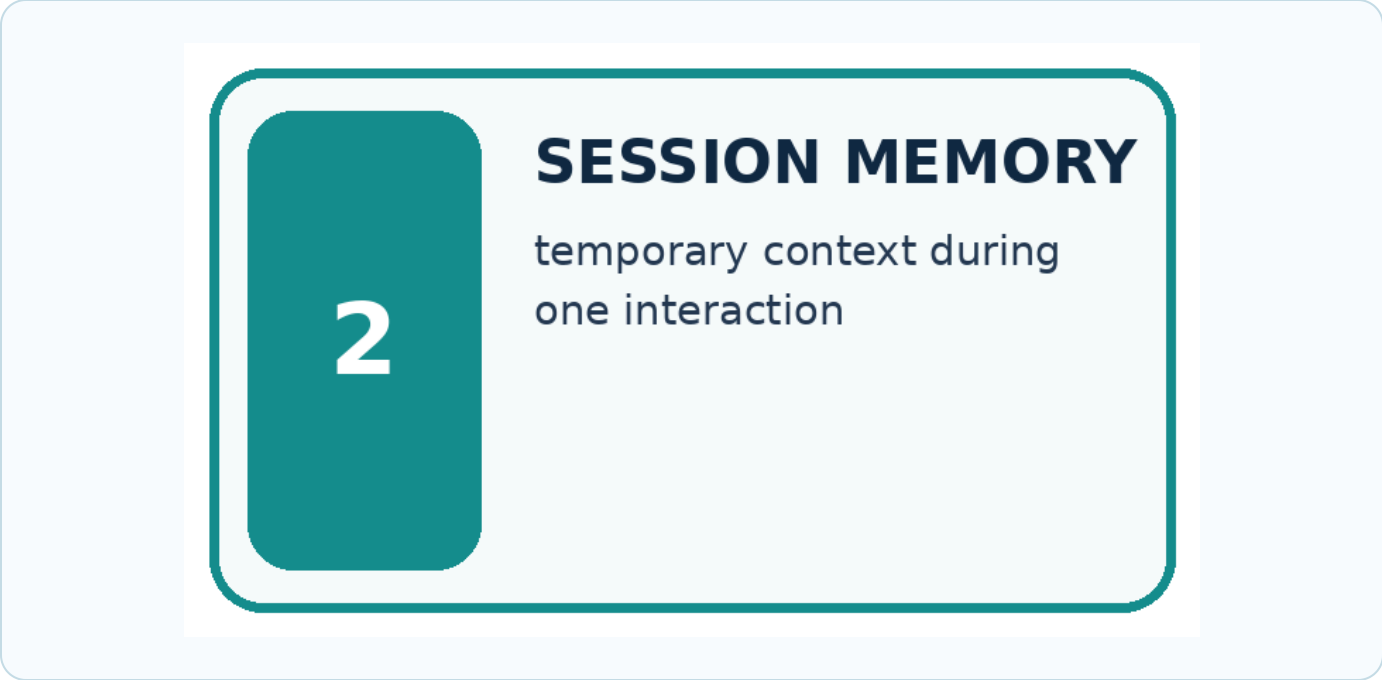
> Use the vocabulary of Conversation State and Memory to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
STATE
information used across turns


2
SESSION MEMORY
temporary context during one interaction







| Word | My own example | Not this / boundary |
|---|---|---|
| state | The current values and information that determine what a process does next. | |
| session memory | Information maintained during the current interaction, without assuming it persists afterwards. | |

| Word | My own example | Not this / boundary |
|---|---|---|
| persistent memory | Information retained beyond the current session by an explicit storage mechanism. | |
| retention rule | A policy specifying what information is kept, for how long and when it is deleted. | |

### Words used in the closer explanation

| Word | Meaning |
|---|---|
| Session state | Information retained within a defined interaction lifetime |
| Correction event | A record updating an earlier active value |

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STUDENT LESSON · 6 OF 18

# Follow a worked example

> Follow the worked trace for Conversation State and Memory, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

**WEEK 20 • WORKED EVIDENCE**

1. Keep current topic and learner’s chosen difficulty for the session.
2. Summarize progress without storing full private conversation.
3. Forget sensitive or unnecessary details.
4. Retrieve approved study notes only when the learner requests that topic.

1. Keep current topic and learner’s chosen difficulty for the session.

2. Summarize progress without storing full private conversation.

3. Forget sensitive or unnecessary details.

4. Retrieve approved study notes only when the learner requests that topic.

Recalculate, retrace, or re**state** the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

Turn 1 selects room A; turn 2 corrects to room B. A later answer uses A from an older summary.

Active **state** should be B under the correction rule. The summary is stale; regenerate or invalidate it and preserve the correction audit within the declared session lifetime.

### Worked explanation · Conversation State and Memory

Compare the intermediate **evidence** with the final **state**ment. Explain why each number or decision follows from the **state**d **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Conversation State and Memory

Keep a reading conversation consistent when the learner changes the selected book.

### The facts we will use

#### Supplied facts

In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/20

# Conversation State and Memory

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. First choice

Store Cedar as this session's selected book.



## 2. Correct choice

Replace Cedar with Maple; do not append an equally active second choice.



## 3. End session

Clear selected_book. A new session begins with no selected book.

**CHANGE ONE THING**

The next session asks "summarise my book" without naming one.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. First choice | Store Cedar as this session’s selected book. |
| 2. Correct choice | Replace Cedar with Maple; do not append an equally active second choice. |
| 3. End session | Clear selected_book. A new session begins with no selected book. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

Conversation **state** is an application record; it is not **evidence** that the model permanently

learned the learner's preferences.

### Keep the stated limit

Read the labelled picture: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Keep the stated limit

Change one thing: The next session asks "summarise my book" without naming one. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Conversation State and Memory claim and explain a repair using the supplied evidence.

### Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

## COMPARE • DIAGNOSE

Weak idea: Remembering everything always creates a better assistant. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

| Evidence record | Expected | Actual | Likely cause / next test |
|---|---|---|---|
| normal case | works | | |
| boundary case | stated rule | | |
| missing input | safe response | | |
| fresh transfer | generalizes | | |

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

**A closer explanation**

### Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

STUDENT LESSON · 8 OF 18

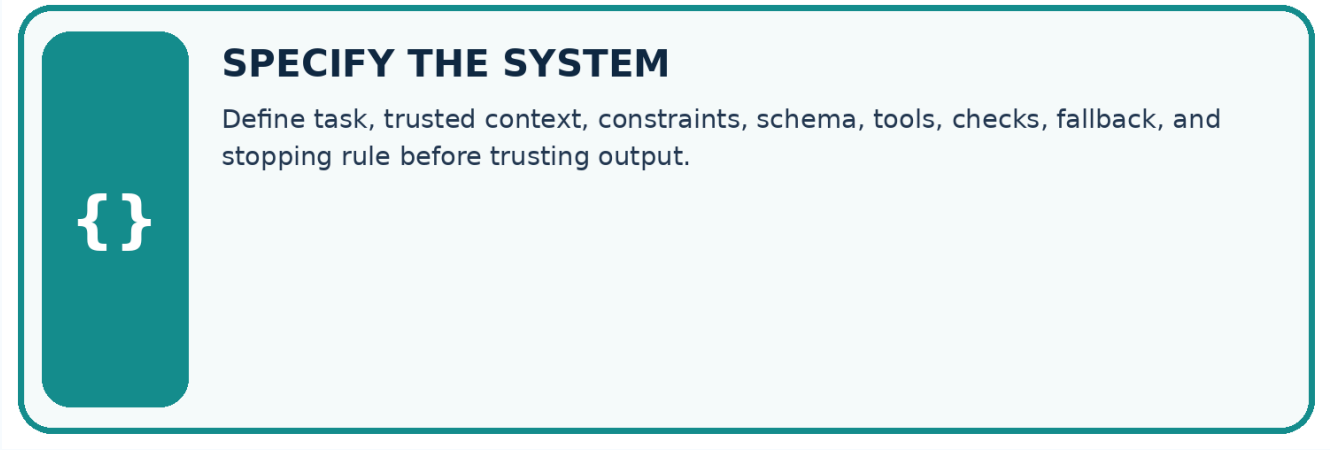
## Practise together

Practise Conversation State and Memory with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.




SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the **state** or chooses a path?

**Write or draw your reasoning and evidence.**

### A closer explanation

Remembered **state**ments are not automatically current or verified; retention should follow the **state**d purpose and policy.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Conversation **state** and retention”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Conversation **state** and retention. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

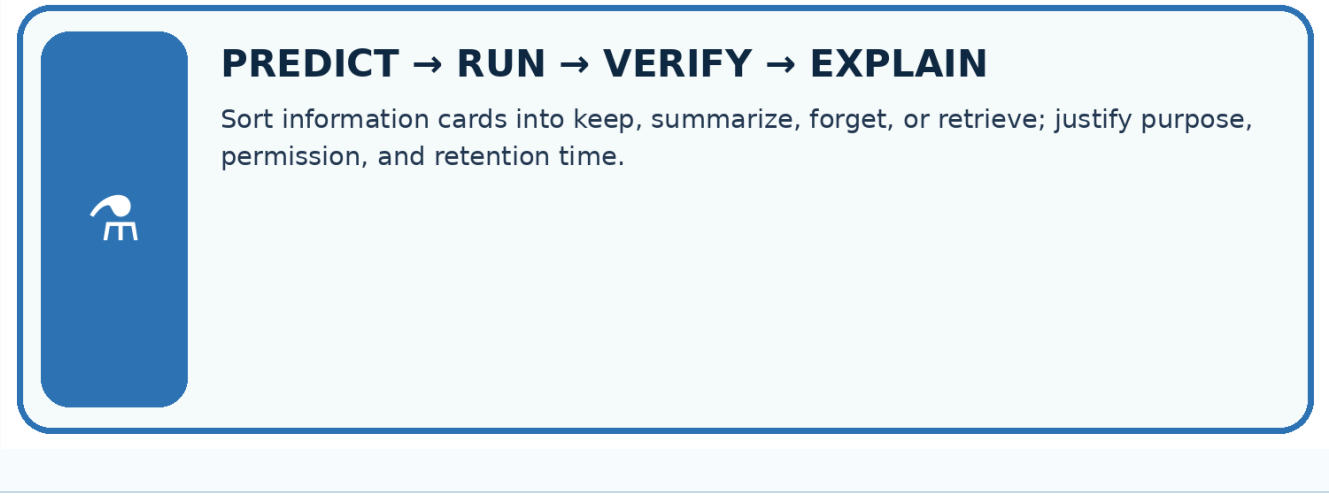
## Run the investigation

> Investigate Conversation State and Memory. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Sort information cards into keep, summarize, forget, or retrieve; justify purpose, permission, and retention time.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

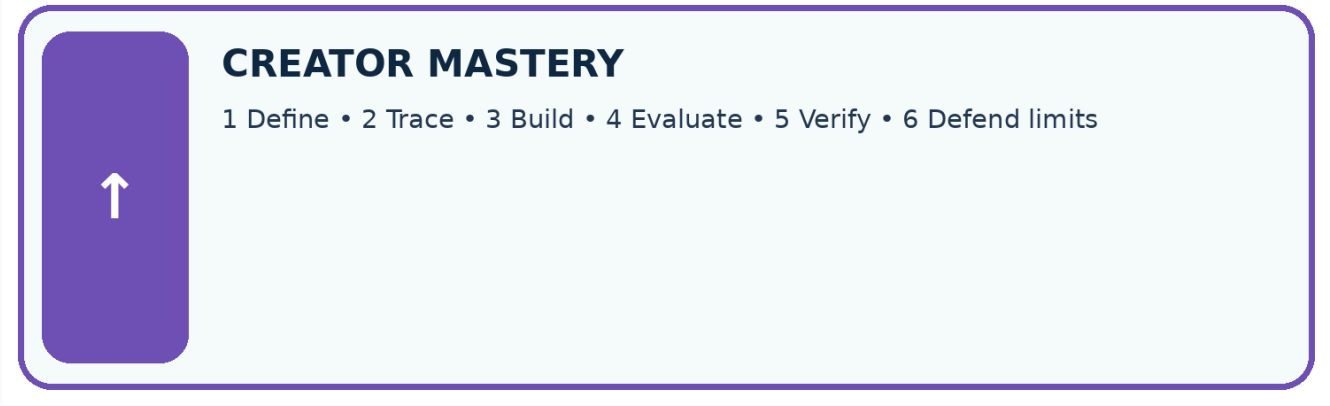
## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Conversation State and Memory. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.




1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, **state**, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

**Write or draw your reasoning and evidence.**

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STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.




THIS WEEK'S CHECKPOINT Store minimal **session state** and make **persistent memory** optional, visible, and erasable.

### Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

What should remain under **human control**?

**Write or draw your reasoning and evidence.**

### Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

**Evidence for your project**

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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STUDENT LESSON · 12 OF 18

## Show what you know

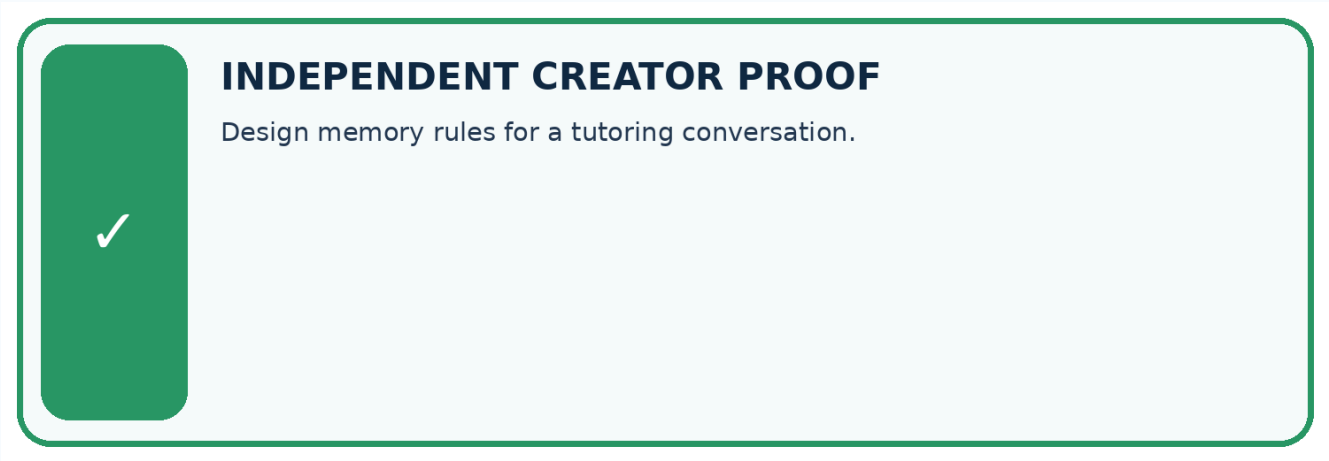
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

### Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.




### Try a more demanding case

CHALLENGE Design memory rules for a tutoring conversation.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

2. My **evidence** that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

### Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

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STUDENT LESSON · 13 OF 18

## Study a complete case

Revisit the complete Conversation State and Memory case for supported practice. Repeated examples do not become fresh assessments.

### Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

**State** stores topic=plants and reading_level=beginner. User requests advanced explanation.

### The question

> What changes?

### Worked explanation

Update reading level, retain topic unless changed. Keep only necessary **state** with correction/reset. Context does not guarantee permanent memory.

### Explain it in your own words

> Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### Trace the supplied case visually

### Update only the state the user changed

| Evidence or stage | Value or result |
| --- | --- |
| Before | topic=plants, level=beginner |

| Evidence or stage | Value or result |
|---|---|
| Request | Use an advanced explanation |
| After | topic=plants, level=advanced |
| Reset | Clear stored fields when requested |

**State** stores topic=plants and reading_level=beginner. User requests advanced explanation.

> What changes?

Update reading level, retain topic unless changed. Keep only necessary **state** with correction/reset. Context does not guarantee permanent memory.

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STUDENT LESSON · 14 OF 18

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Apply a new case independently

A session stores lab L1 then corrects to L2; session ends. A new session has no room **input**. **State** the active values at each stage.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A learner corrects their project topic from oceans to forests. The next reply still uses oceans. What **state** should change, and what should not be saved unnecessarily?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

### Connect to your project

Store minimal **session state** and make **persistent memory** optional, visible, and erasable.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

### Transfer practice from the original programme

The conversation **state** is topic=plants, level=Grade 8. The user says 'Keep the topic, explain for Grade 6'. Write the updated **state** and a reset behaviour.

**Explain the mechanism, evidence and one limit.**

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

Use this map to navigate Conversation State and Memory, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Keep a reading conversation consistent when the learner changes the selected book.

| By the end, I can | How I will show it |
| --- | --- |
| Explain the mechanism and terminology of conversation state and retention | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Conversation State and Memory without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of conversation **state** and retention

**My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep a reading conversation consistent when the learner changes the selected book.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? Clear selected_book. A new session begins with no selected book.

1. First choice

2. Correct choice

3. End session

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: End session Clear selected_book. A new session begins with no selected book. Read the picture in order. Clear selected_book.

A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The next session asks “summarise my book” without naming one. Which conclusion is supported by the supplied facts?

1. Ask which book the learner wants.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Ask which book the learner wants. The declared policy clears **session state**, so yesterday's choice is unavailable.

**Repair a missing building block**

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

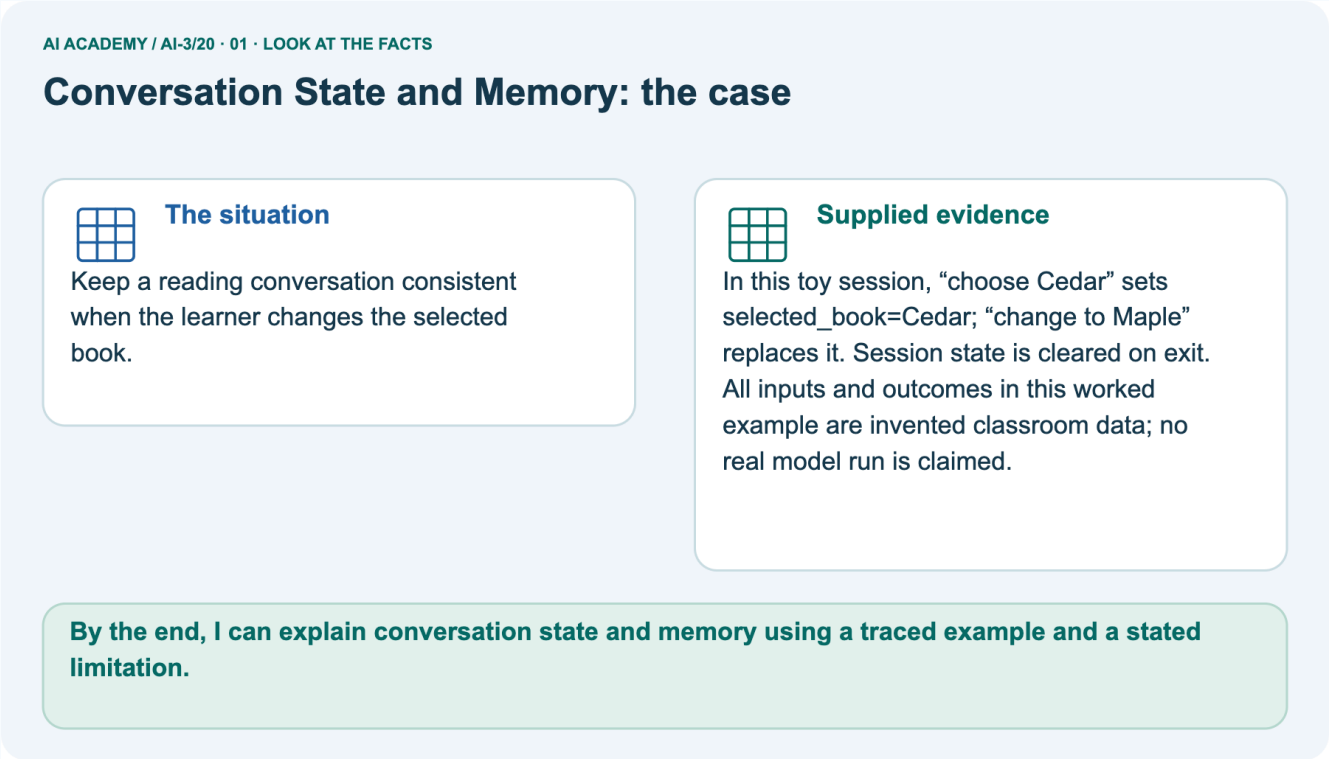
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**STUDENT LESSON · 18 OF 18**

## Visual concept journey

> Explain each Conversation State and Memory graphic using its labels, deciding evidence and relevant limitation.

## 01. Conversation State and Memory: the case



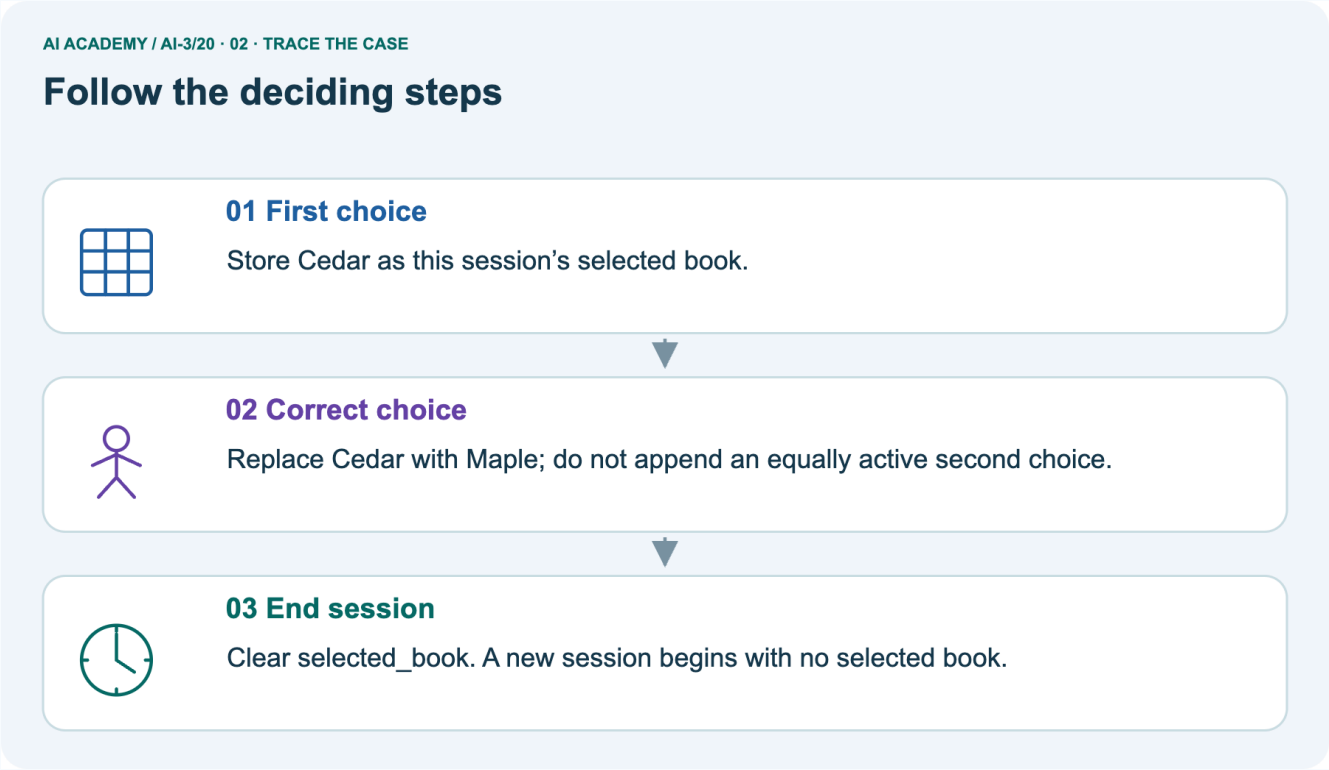

Follow the labelled picture with the matching explanation. Open the vector diagram

Keep a reading conversation consistent when the learner changes the selected book. Supplied facts: In this toy session, “choose Cedar” sets selected_book=Cedar; “change to Maple” replaces it. **Session state** is cleared on exit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps



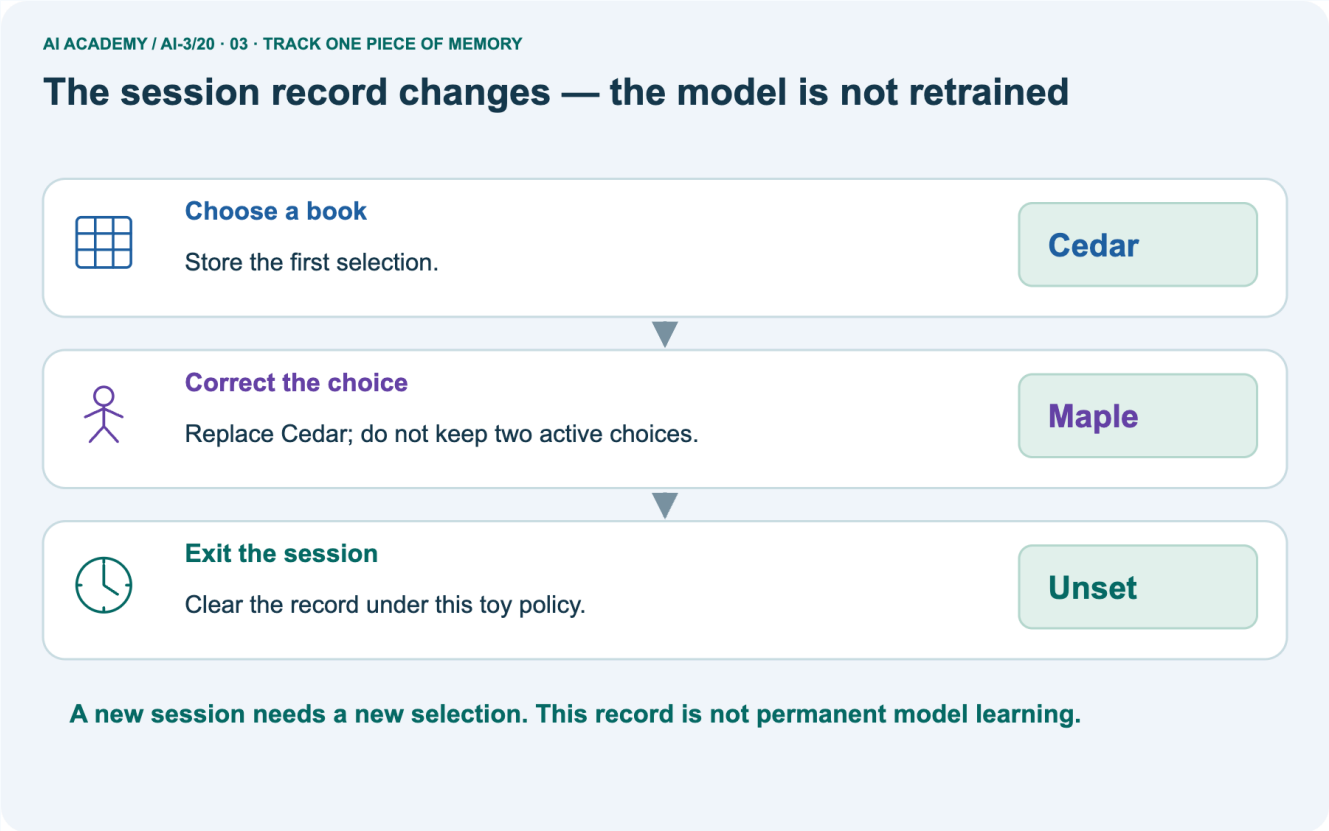

Follow the labelled picture with the matching explanation. Open the vector diagram

First choice: Store Cedar as this session's selected book. → Correct choice: Replace Cedar with Maple; do not append an equally active second choice. → End session: Clear selected_book.

A new session begins with no selected book.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. The session record changes — the model is not retrained

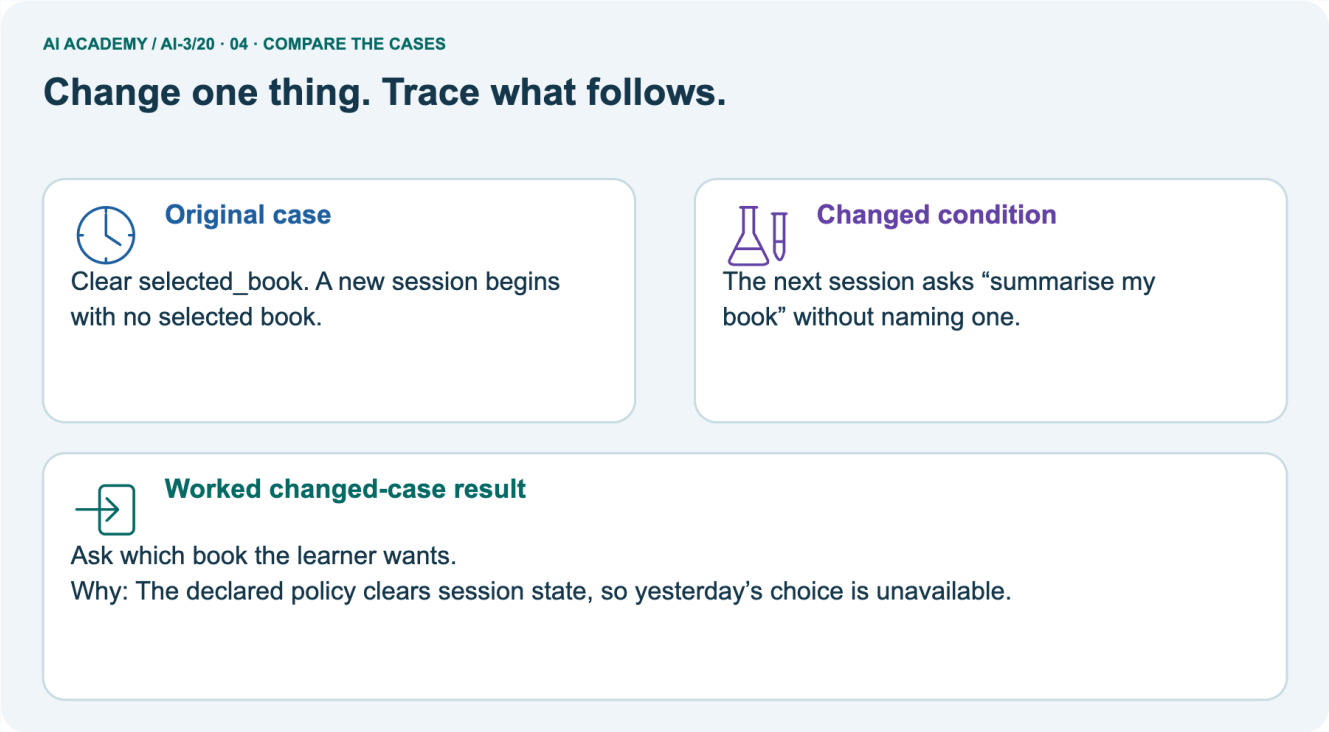



Follow the labelled picture with the matching explanation. Open the vector diagram

The selected_book **session state** is Cedar after the first choice, Maple after correction, and unset after exit. This is an application record, not proof of permanent learning.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



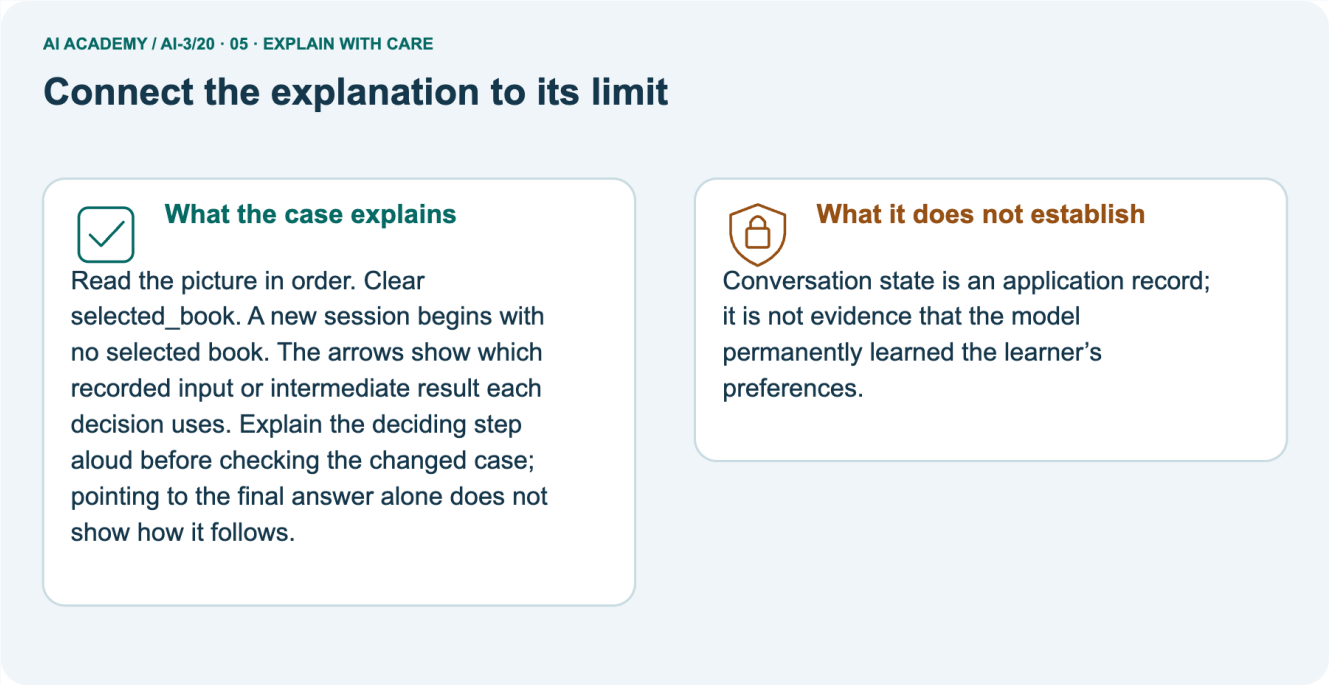

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Clear selected_book. A new session begins with no selected book. Changed condition: The next session asks “summarise my book” without naming one. Expected result: Ask which book the learner wants.

Reason: The declared policy clears **session state**, so yesterday’s choice is unavailable.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

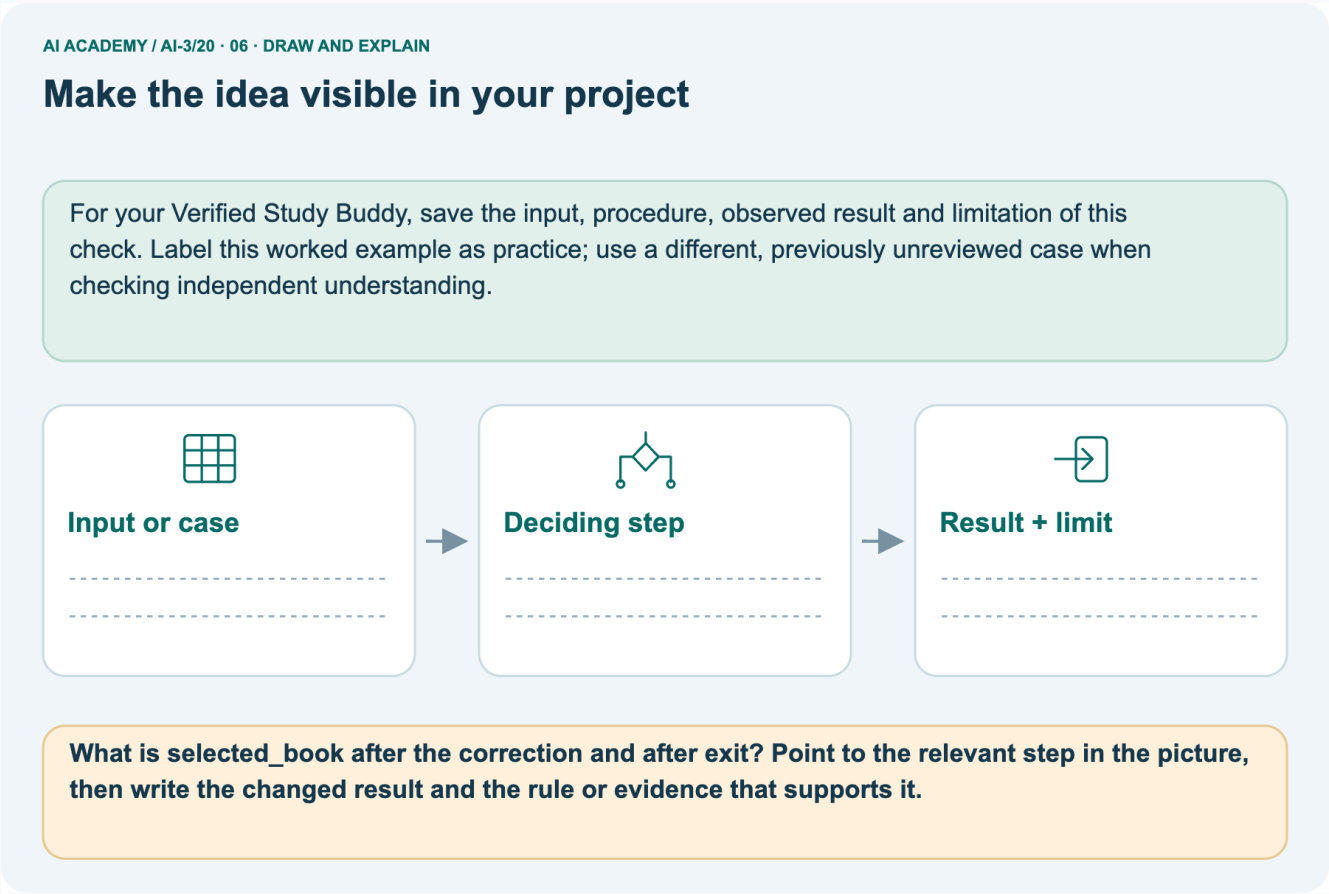
Read the picture in order. Clear selected_book. A new session begins with no selected book. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Conversation **state** is an application record; it is not **evidence** that the model permanently learned the learner's preferences.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.